

Online Learning of Control-Oriented Lateral Tire-Force Maps for Vehicle MPC

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Abstract—Vehicle model predictive control (MPC) repeatedly evaluates a tire-force model at predicted slip angles over its prediction horizon. Hence, an instantaneous force correction is insufficient: the learned object should be a reusable slip-angle-indexed map. This paper presents an online method that updates front and rear lateral tire-force maps from lateral-yaw state errors without direct tire-force measurements. A neural identifier supplies the instantaneous adaptation signal, while a predefined slip-angle table retains the learned relation over the previously excited domain. CarMaker simulations with changing tire characteristics and steering transitions show that the proposed method reduces trajectory force errors and frozen-horizon prediction errors compared with instantaneous and recent-window updates. The resulting compact model can be directly evaluated inside a vehicle MPC prediction horizon.

Index Terms—Vehicle dynamics, model predictive control, online learning, tire-force estimation, neural identifier.

I. INTRODUCTION

The accuracy of vehicle MPC depends strongly on the tire-force model used to propagate lateral velocity and yaw rate. Conventional analytical or learned tire models provide an explicit relation between slip angle and force, but an offline model may become inaccurate when friction, load, temperature, or tire condition changes. Online residual-learning methods can reduce the current vehicle-response mismatch [1], [2], and vehicle-response-based methods can update tire models without force sensors [3], [4]. However, a correction that fits only the current trajectory does not necessarily remain consistent at the other slip angles visited by an MPC prediction horizon.

This work treats online tire-force learning as formation of a control-oriented prediction map. The main idea is to combine (i) a neural identifier driven by the lateral-yaw state error and (ii) a slip-angle-domain memory that preserves the learned map at previously excited points. The method does not aim to recover the globally exact tire physics. Instead, it maintains a compact and differentiable force map over the domain supported by the observed maneuver.

II. PREDICTION MODEL

Consider the two-degree-of-freedom lateral-yaw model with $x = [v_y, r]^T$:

$$\dot{x} = f_0(x, v_x) + B_f F_{y,f} + B_r F_{y,r}, \quad f_0 = \begin{bmatrix} -v_x r \\ 0 \end{bmatrix}, \quad (1)$$

$$B_f = \begin{bmatrix} 1/m \\ l_f/I_z \end{bmatrix}, \quad B_r = \begin{bmatrix} 1/m \\ -l_r/I_z \end{bmatrix}. \quad (2)$$

The front and rear slip angles are

$$\alpha_f = \delta_f - \tan^{-1} \frac{v_y + l_f r}{v_x}, \quad \alpha_r = \delta_r - \tan^{-1} \frac{v_y - l_r r}{v_x}. \quad (3)$$

For $i \in \{f, r\}$, the learned axle-force map is parameterized by fixed nonlinear basis functions,

$$\begin{aligned} \hat{F}_i(\alpha; \theta_i) &= \theta_i^\top \phi_i(\alpha), \\ \phi_{i,j}(\alpha) &= \tanh(\lambda_{i,j}(\alpha - b_{i,j})) - \tanh(-\lambda_{i,j} b_{i,j}). \end{aligned} \quad (4)$$

The shift enforces $\hat{F}_i(0) = 0$, and the linear-in-parameter form is inexpensive and differentiable. At control time k , the weights are frozen over the MPC horizon:

$$\hat{F}_{i,k+j|k} = \hat{F}_i(\alpha_{i,k+j|k}; \theta_{i,k}), \quad j = 0, \dots, N-1. \quad (5)$$

Thus, unlike a time-indexed adaptive correction, the same learned function is evaluated at all predicted slip angles.

III. ONLINE TIRE-FORCE MAP UPDATE

The neural identifier uses measured or estimated states and the learned front/rear force maps:

$$\dot{\hat{x}} = f_0(x, v_x) + B_f \hat{F}_f + B_r \hat{F}_r + A_e(\hat{x} - x), \quad (6)$$

where A_e is Hurwitz. With $e = x - \hat{x}$, the error dynamics become

$$\dot{e} = A_e e + B_f(F_{y,f} - \hat{F}_f) + B_r(F_{y,r} - \hat{F}_r). \quad (7)$$

Let $P = P^\top > 0$ solve $A_e^\top P + P A_e = -Q$, $Q > 0$. The cross term in the derivative of $V = e^\top P e / 2$ provides the instantaneous adaptation direction $\phi_i(\alpha_i) B_i^\top P e$. It uses only the state error and known vehicle parameters, rather than a measured tire-force label.

Instantaneous adaptation acts only near the current slip angle. To avoid losing the relation learned elsewhere, predefined breakpoints $\{\alpha_{i,j}^p\}_{j=1}^{N_p}$ are activated when the trajectory crosses them. Once the identifier error is sufficiently small, the current model output is stored as $F_{i,j}^p$. With the basis matrix Φ_i^p , active-point mask M_i^p , and number of active points n_i^p , the consistency loss is

$$J_{\text{tab},i} = \frac{1}{2n_i^p} \left\| (M_i^p)^{1/2} (\Phi_i^p \theta_i - F_i^p) \right\|^2. \quad (8)$$

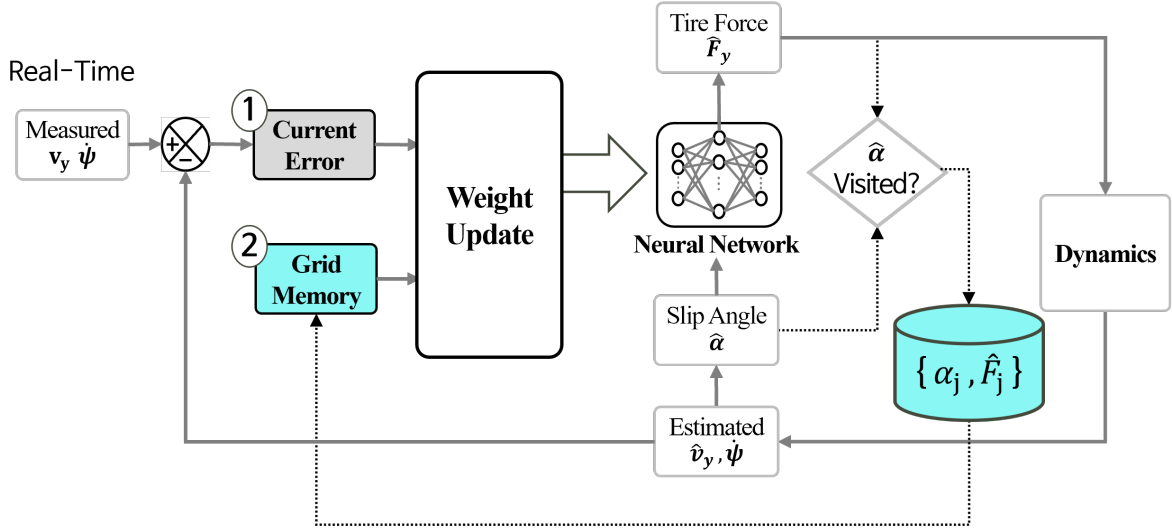


Fig. 1. Online learning architecture. The identifier error adapts the current force map, and the slip-angle table retains previously learned map values for reuse in the MPC prediction horizon.

The complete update combines the current state-error signal, table consistency, and leakage:

$$\dot{\theta}_i = \eta_{\text{inst}} \phi_i(\alpha_i) B_i^\top P e - \eta_{\text{tab}} \nabla_{\theta_i} J_{\text{tab},i} - \eta_{\text{reg}} \theta_i. \quad (9)$$

The first term adapts to a changed tire characteristic, while the second limits unintended drift at previously excited slip angles. The table stores model outputs rather than measured forces and therefore introduces no additional force sensor.

A. Lyapunov Analysis of the Adaptive Identifier

The following Lyapunov argument establishes boundedness of the identifier and adaptive parameters; it is not a closed-loop MPC stability proof. For the proposed method, let $\eta_{\text{inst}}, \eta_{\text{tab}}, \eta_{\text{reg}} > 0$, and assume operation on a compact domain with $v_x \geq v_{\min} > 0$, bounded basis functions and derivatives, and bounded force-approximation, table-data, and slip-reconstruction errors. Let θ_i^* be a constant best parameter over the visited domain, $\tilde{\theta}_i = \theta_i^* - \theta_i$, and write $F_{y,i} = (\theta_i^*)^\top \phi_i + \epsilon_i$. After a finite activation time T_g , assume the normalized active-table Gramian satisfies

$$\lambda_{\min} \left(\frac{1}{n_i^p} (\Phi_i^p)^\top M_i^p \Phi_i^p \right) \geq \underline{\Delta}_{p,i} > 0. \quad (10)$$

This is a visited-domain excitation condition, rather than global tire-force identifiability. Consider

$$V = \frac{1}{2} e^\top P e + \frac{1}{2\eta_{\text{inst}}} \sum_{i \in \{f,r\}} \|\tilde{\theta}_i\|^2. \quad (11)$$

Using (7) and (9), the instantaneous update cancels the state-parameter cross term, giving

$$\begin{aligned} \dot{V} = & -\frac{1}{2} e^\top Q e + \sum_i e^\top P B_i \epsilon_i + e^\top P d_\alpha \\ & + \frac{\eta_{\text{tab}}}{\eta_{\text{inst}}} \sum_i \tilde{\theta}_i^\top \nabla_{\theta_i} J_{\text{tab},i} + \frac{\eta_{\text{reg}}}{\eta_{\text{inst}}} \sum_i \tilde{\theta}_i^\top \theta_i, \end{aligned} \quad (12)$$

where d_α denotes the bounded slip-reconstruction mismatch. If $F_i^p = \Phi_i^p \theta_i^* + \delta_i^p$, the table term contains

$$-\frac{1}{n_i^p} \tilde{\theta}_i^\top (\Phi_i^p)^\top M_i^p \Phi_i^p \tilde{\theta}_i - \frac{1}{n_i^p} \tilde{\theta}_i^\top (\Phi_i^p)^\top M_i^p \delta_i^p. \quad (13)$$

Equation (10) and Young's inequality bound this by a negative quadratic in $\tilde{\theta}_i$ plus a bounded table-data residual. Similarly, $\tilde{\theta}_i^\top \theta_i \leq -\|\tilde{\theta}_i\|^2/2 + \|\theta_i^*\|^2/2$, while the approximation and reconstruction terms are bounded by a sufficiently small multiple of $\|e\|^2$ plus a constant. Therefore, for positive constants $c_e, c_{\theta,i}$ and finite d ,

$$\dot{V} \leq -c_e \|e\|^2 - \sum_i c_{\theta,i} \|\tilde{\theta}_i\|^2 + d \leq -aV + b, \quad (14)$$

with $a, b > 0$. Hence $V(t) \leq e^{-a(t-T_g)} V(T_g) + (b/a)(1 - e^{-a(t-T_g)})$; the identifier error and parameter errors are uniformly ultimately bounded under the stated assumptions.

IV. SIMULATION RESULTS

The method was evaluated using open-loop CarMaker data from a BMW passenger-vehicle model at approximately 60 km/h. The model parameters were $m = 1600$ kg, $I_z = 2986.624$ kg m², $l_f = 1.488$ m, and $l_r = 1.487$ m. The common neural-model settings and all method-specific learning gains are reported in Table I. Three steering cases were considered: C1 changed the effective tire characteristic during a sine maneuver; C2 changed from a large to a small-amplitude sine; and C3 changed from a sine to a step steer.

The proposed method was compared with Baseline1, which uses only the instantaneous term in (9), and Baseline2, which minimizes a recent-window one-step prediction error.

A. Consistency Over the Excited Slip-Angle Domain

Case C2 evaluates whether a relation learned during the initial large-amplitude sine is retained after the maneuver excites only a narrow slip-angle region. This matters because

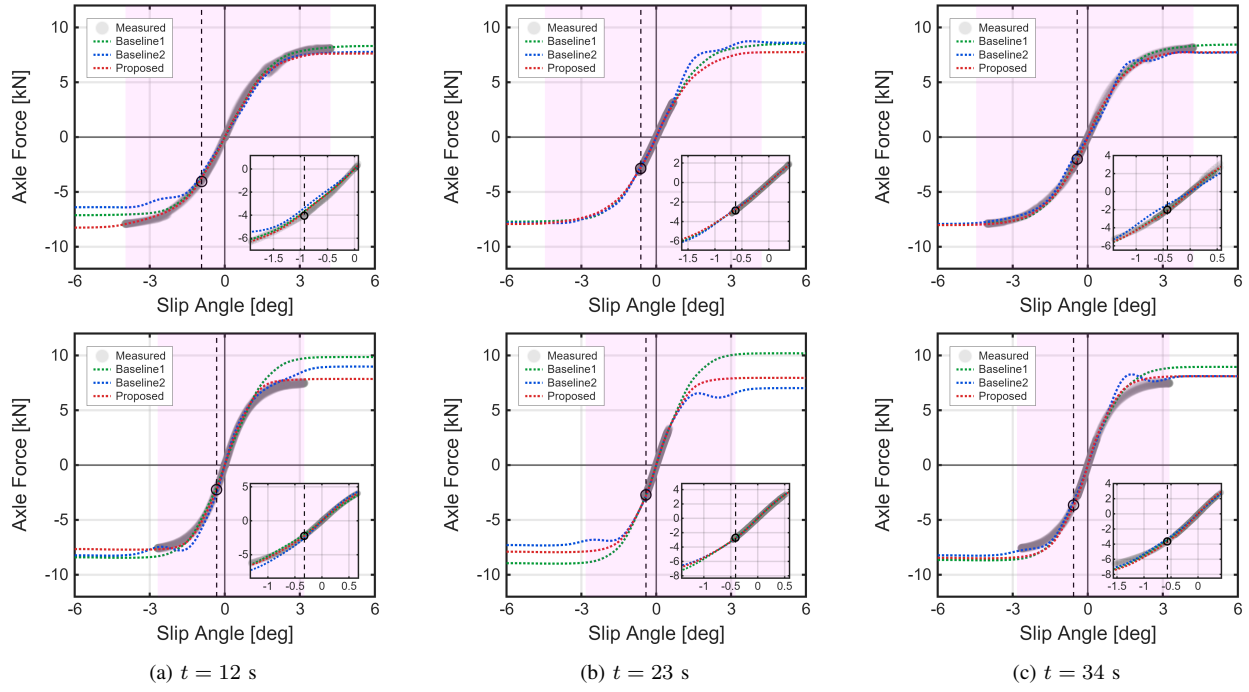


Fig. 2. Learned tire-force maps for C2. The top and bottom rows show the front and rear axles, respectively; the columns correspond to $t = 12, 23$, and 34 s. Gray samples denote measured trajectory points, and the shaded region denotes the slip-angle range excited up to each snapshot.

TABLE I
NEURAL MODEL AND METHOD-SPECIFIC LEARNING SETTINGS

Entry	Type	Main settings
Common	neural model	$T_s = 0.01$ s; tanh centers $-8 : 1 : 8$ deg; table points $-10 : 0.5 : 10$ deg; $\beta_f, \beta_r = 100 \text{ rad}^{-1}$; $v_{\min} = 0.1$ m/s; force scale 10^4 N; initial weight scale 10^{-3} ; seed 1234
Baseline1	instantaneous	$\eta_{\text{inst}} = 1.0, \eta_{\text{tab}} = 0, \eta_{\text{reg}} = 5 \times 10^{-3}$
Baseline2	windowed	$K = 300, \eta_{\text{mb}} = 1.0, \gamma_{\text{mb}} = 0.9,$
	mini-batch	$\lambda_{\text{mb}} = 10^{-4}, W_{\text{mb}} = I$
Proposed	instantaneous + table	$\eta_{\text{inst}} = 2.0, \eta_{\text{tab}} = 3.0, \eta_{\text{reg}} = 10^{-3}, \gamma_F = 0.1, \epsilon_{\text{tab}} = 0.05$

TABLE II
TRAJECTORY FORCE RMSE IN THE POST-TRANSITION INTERVAL

Case	Method	RMSE($F_{y,f}$) [kN]	RMSE($F_{y,r}$) [kN]
C2	Baseline1	0.254	0.214
	Baseline2	0.322	0.258
	Proposed	0.236	0.208
C3	Baseline1	0.274	0.293
	Baseline2	0.278	0.291
	Proposed	0.271	0.293

an MPC prediction can revisit a broader, previously excited domain even when the current operating point is local. Figure 2 preserves the three snapshots used in the full manuscript. At $t = 12$ s, the three methods are still learning from broad excitation. Around $t = 23$ s, their curves are similar near the current slip angle but differ outside the local region. At $t = 34$ s, the proposed table term retains the previously stored shape more consistently than the instantaneous or recent-window updates. The C2 force RMSE in Table II gives the same conclusion along the realized trajectory.

TABLE III
ADAPTATION UNDER TIME-VARYING TIRE CHARACTERISTIC

Method	20–42 s		42–50 s	
	Front [kN]	Rear [kN]	Front [kN]	Rear [kN]
Baseline1	0.187	0.172	0.218	0.197
Baseline2	0.642	0.178	0.296	0.198
Proposed	0.184	0.161	0.207	0.185

B. Adaptation to a Time-Varying Tire Characteristic

Case C1 changes the effective lateral-force scaling during the sine maneuver. Figure 3 shows that the early learned curves are close because the characteristic has only begun to change. By $t = 38$ s, all methods have reacted to the new response, but their unexcited-domain behavior differs. The proposed method follows the changed characteristic while suppressing unnecessary deformation of the curve over the excited interval. In the late 42–50 s interval, its front and rear force RMSE values are 0.207 and 0.185 kN, respectively, compared with 0.218/0.197 kN for Baseline1 and 0.296/0.198 kN for Baseline2. Table III preserves the original changing-interval and late-interval comparison.

C. Frozen-Horizon Prediction Performance

For MPC-oriented validation, the weights were frozen at each rollout start and the model was propagated for 1.0 s using the future open-loop steering and speed sequence. Table IV retains the prediction errors as numerical results rather than an additional figure. The proposed method gives the smallest v_y and r prediction errors in all three cases, supporting reuse of the learned map beyond the current operating point.

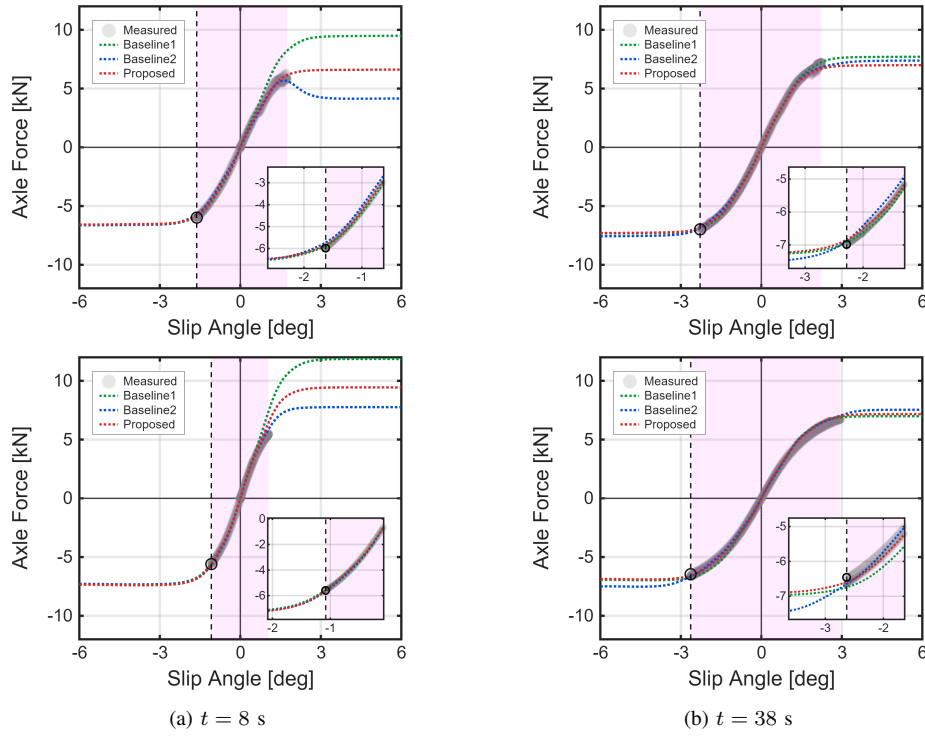


Fig. 3. Learned tire-force maps for the time-varying tire-characteristic case C1. The top and bottom rows show the front and rear axles, respectively; the columns correspond to $t = 8$ and 38 s. Gray samples denote measured trajectory points, and the shaded region denotes the slip-angle range excited up to each snapshot.

TABLE IV
FROZEN-HORIZON PREDICTION ERRORS

Case	Method	0.5 s		1.0 s	
		RMSE(v_y)	RMSE(r)	RMSE(v_y)	RMSE(r)
C1	Baseline1	0.2589	0.0541	1.0670	0.1494
	Baseline2	0.3791	0.0893	1.4929	0.2215
	Proposed	0.2216	0.0474	0.9386	0.1297
C2	Baseline1	0.3726	0.0777	1.4898	0.1882
	Baseline2	0.4535	0.0954	1.6772	0.2125
	Proposed	0.3101	0.0644	1.2789	0.1648
C3	Baseline1	0.3698	0.0769	1.4876	0.1875
	Baseline2	0.4391	0.0967	1.6715	0.2179
	Proposed	0.3094	0.0643	1.2773	0.1658

V. CONCLUSION

An online learning method was presented for constructing slip-angle-indexed lateral tire-force maps for vehicle MPC. A neural identifier adapts the maps from lateral–yaw state errors, and a predefined slip-angle table preserves consistency over the excited domain. The identifier and parameter errors are uniformly ultimately bounded under the stated visited-domain assumptions. CarMaker results indicate improved force-map retention and frozen-horizon prediction relative to instantaneous and recent-window baselines. Future work will integrate the learned map into closed-loop MPC and validate it on a test vehicle.

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